Seminar 7.
15 aprilie 2014

1. Să se calculeze

$$i_2 = \int_{5\pi}^{7\pi} (1-\sin x)^{99} \sin 99x \, dx$$

$-\pi \quad 0 \quad \pi \quad 2\pi \quad 3\pi \quad 4\pi \quad 5\pi \quad 6\pi \quad 7\pi$

$$= -\int_{0}^{2\pi} (1+\sin x)^{99} \sin 99x \, dx$$

$$\sin x = \frac{e^{jx} - e^{-jx}}{2} = \frac{e^{2jx}-1}{2je^{jx}}$$

$\boxed{e^{jx} = z} \Rightarrow |z| = 1$ $\qquad \int_{a}^{a+2\pi} \longrightarrow \int_{|z|=1}$

$\Downarrow$

$$dz = je^{jx}\, dx$$

$$dx = \frac{dz}{jz}$$

$$i_1 = \int_{5\pi}^{7\pi} (1-\sin x)^{99} \cos 99x \, dx$$

$$i = i_1 + j i_2$$

$$= \int_{5\pi}^{7\pi} (1-\sin x)^{99} (\cos 99x + j \sin 99x)\, dx$$

$$= \int_{5\pi}^{7\pi} (1-\sin x)^{99} e^{99jx} \, dx$$

$$= \int_{|z|=1} \left(1 - \frac{z^2-1}{2jz}\right)^{99} \cancel{z^{99}} \frac{dz}{jz}$$

$$= \int_{|z|=1} \frac{(2jz - z^2 + 1)^{99}}{2^{99} \cdot j^{99} \cdot \cancel{z^{99}}} \cdot \cancel{z^{99}} \cdot \frac{dz}{jz}$$

$$i = \frac{1}{2^{99}} \int\limits_{|z|=1} \frac{(2jz - z^2+1)^{99}}{z} dz$$

$z_0 = 0$ – pol simplu

$$= \frac{1}{2^{99}} 2\pi j \operatorname{Rez}(f; 0)$$

$$= \frac{1}{2^{99}} 2\pi j \left. (2jz - z^2+1)^{99} \right|_{z=0}$$

$$= \frac{1}{2^{98}} \pi j = i_1 + j i_2$$

$$\Rightarrow i_1 = 0$$

$$i_2 = \frac{1}{2^{98}} \pi$$

2. $$\int_{-\pi}^{3\pi} (1+\cos x)^{2n} \cos nx \, dx = 2 \int_{-\pi}^{\pi} (1+\cos x)^{2n} \cdot \cos nx \, dx$$

$e^{jx} = z$

$$\cos x = \frac{z^2+1}{2z}$$

$$\sin x = \frac{z^2-1}{2jz}$$

$|z| = 1$

$$dx = \frac{dz}{zj}$$

$i = i_1 + j i_2$

$$i_1 = \int_{-\pi}^{\pi} (1+\cos x)^{2n} \cos nx \, dx \qquad i_2 = \int_{-\pi}^{\pi} (1+\cos x)^{2n} \sin nx \, dx$$

$$i = \int_{-\pi}^{\pi} (1+\cos x)^{2n} e^{jnx} \, dx$$

$$= \oint_{|z|=1} \left(1 + \frac{z^2+1}{2z}\right)^{2n} \cdot z^n \cdot \frac{dz}{jz}$$

$$= \oint_{|z|=1} \frac{(z+1)^{4n}}{2^{2n} \cdot \cancel{z^{2n}}} \cdot \cancel{z^n} \, \frac{dz}{jz}$$

$$= \frac{1}{j2^{2n}} \oint \frac{(z+1)^{4n}}{z^{n+1}} \, dz$$

$$= \frac{1}{2^{2n} j} \cdot 2\pi j \; \mathrm{Rez}(f, 0)$$

$$\frac{(z+1)^{4n}}{z^{n+1}} = \frac{1}{z^{n+1}} \sum_{k=0}^{4n} \binom{4n}{k} z^{4n-k} \cdot 1^k$$

$$= \sum_{k=0}^{4n} \binom{4n}{k} z^{4n-k-n-1}$$

$$= \sum_{k=0}^{4n} \binom{4n}{k} z^{3n-k-1}$$

$$k = 3n$$

$$\Rightarrow a_{-1} = \binom{4n}{3n} = \binom{4n}{n}$$

$$\mathrm{Rez}(f, 0) = \frac{(4n)!}{(3n)! \cdot n!} =$$

$$i = \frac{1}{2^{2n} j} \cdot 2\pi j \cdot \binom{4n}{n}$$

$$\frac{\pi}{2^{2n-1}}\binom{4n}{n} = I_1 + j I_2$$

$$\Rightarrow I_2 = 0 \Rightarrow I_1 = \frac{\pi}{2^{2n-1}}\binom{4n}{n}$$

dar $$\int_{-\pi}^{3\pi} (1+\cos x)^{2n} \cos nx \, dx = 2 \cdot \frac{\pi}{2^{2n-1}}\binom{4n}{n}$$

$$= \frac{\pi}{2^{2n-2}}\binom{4n}{n}$$

3° $\sin z - \cos z = 2$

$e^{jz} = t$

$$\frac{t^2-1}{2jt} - \frac{t^2+1}{2t} = 2 \quad \Big| \, 2jt$$

$$t^2 - 1 - jt^2 - j = 4jt$$

$$t^2(1-j) - 4jt - (1+j) = 0$$

$$\Delta = -8$$

$$t_{1,2} = \frac{4j \pm 2j\sqrt{2}}{2(1-j)} = \frac{j(2 \pm \sqrt{2})}{1-j}$$

$$= \underbrace{\frac{(2 \pm \sqrt{2})}{2}}_{>0}(j-1)$$

$$e^{jz} = \frac{2+\sqrt{2}}{2}(-1+j) \quad \Big| \, \mathrm{Ln}$$

$$jz = \mathrm{Ln}\left(\frac{2+\sqrt{2}}{2}(-1+j)\right) = \ln\left|\frac{2+\sqrt{2}}{2}(-1+j)\right| + j\arg\left[\frac{2+\sqrt{2}}{2}(-1+j)\right] + 2k\pi$$

$$= \frac{2+\sqrt{2}}{2}\underbrace{\ln|-1+j|}_{\sqrt{2}} + j\left(\underbrace{\arg\frac{2+\sqrt{2}}{2}}_{=0} + \underbrace{\arg(-1+j)}_{3\pi/4} + 2k\pi\right)$$

$$= -\frac{2+\sqrt{2}}{4}\ln 2 + j\left(\frac{3\pi}{4} + 2k_1\pi\right) \quad \Big| : j$$

$$\Rightarrow z_1 = \left(\frac{3\pi}{4} + 2k_1\pi\right) - j\,\frac{(2+\sqrt{2})}{4}\ln 2$$

$$z_2 = \left(\frac{3\pi}{4} + 2k_2\pi\right) - j\,\frac{2-\sqrt{2}}{4}\ln 2$$

4°. $|\sin z| \cancel{\leq} 1$

$z \in \mathbb{C}$

$$\sin z = \sin(x + jy)$$

$$= \sin x \cos jy + \cos x \cdot \sin jy = \sin x\, ch\, y + j \cos x\, sh\, y$$

$$\cos jy = \frac{e^{jjy} + e^{-j \cdot jy}}{2} = \frac{e^{-y} + e^{y}}{2} = ch\, y$$

$$\sin jy = \frac{e^{jjy} - e^{-jjy}}{2j} = -\frac{e^{-y} - e^{y}}{2j} = j\, sh\, y$$

$$|\sin z| = \sqrt{\underbrace{\sin^2 x}_{1-\cos^2 x} \cdot ch^2 y + \cos^2 x\, sh^2 y}$$

$$\boxed{ch^2 y - sh^2 y = 1}$$

$$= \sqrt{ch^2 y - \cos^2 x\, ch^2 y + \cos^2 x\, sh^2 y}$$

$$= \sqrt{ch^2 y - \cos^2 x\,(ch^2 y - sh^2 y)}$$

$$= \sqrt{ch^2 y - \cos^2 x}$$

$$\boxed{ch\, y \geq 1}$$

5°. $Z = \frac{z+j}{jz+3} \; (=f(z))$

$$Z' = \frac{jz+3 - j(z+j)}{(jz+3)^2}$$

$$= \frac{\cancel{jz}+3-\cancel{jz}+1}{(jz+3)^2}$$

$$= \frac{4}{(jz+3)^2}$$

$$\boxed{|Z'| = 1}$$

$$\Rightarrow \left|\frac{4}{(jz+3)^2}\right| = 1$$

$$\frac{|jz+3|^2}{4} = 1$$

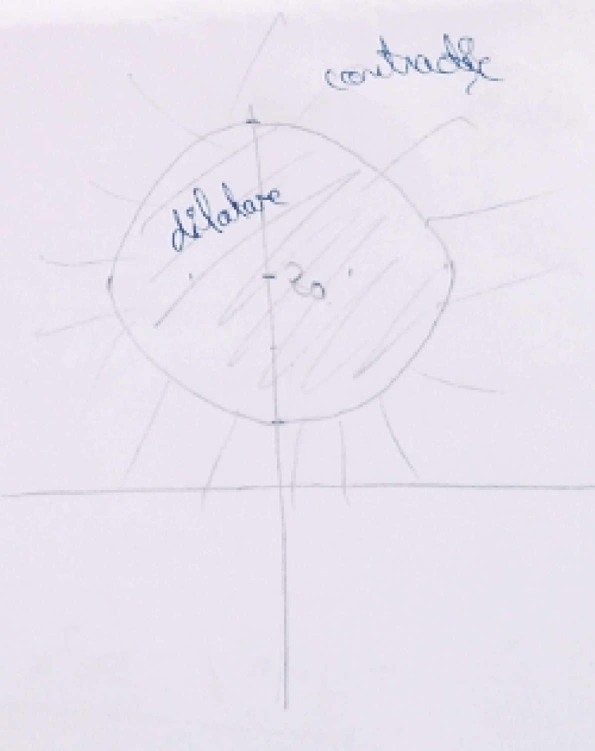
$$|jz+3|^2 = 4$$

$$|jz+3| = 2$$

$$|j(z-3j)| = 2$$

$$|j|\,|z-3j| = 2$$

$$|z-3j| = 2$$



$$\left|\frac{4}{(jz+3)^2}\right| > 1 \Rightarrow \frac{|jz+3|^2}{2} < 1 \Rightarrow |z-3j| < 2$$

6°. $Z = \frac{2z - j + 2}{j - z - 1}$

$$Z' = \frac{2(j - z - 1) + (2z - j + 2)}{(j - z - 1)^2}$$

$$= \frac{2j - \cancel{2z} - \cancel{2} + \cancel{2z} - j + \cancel{2}}{(j - z - 1)^2}$$

$$Z' = \frac{j}{(j - z - 1)^2}$$

$$\arg Z' = \arg j - \arg (j - z - 1)^2 = 0$$

$$= \frac{\pi}{2} - 2\arg(j - z - 1) = 0$$

$$\arg(j - z - 1) = \frac{\pi}{4}$$

$z = x + jy$

$$\arg(j - x - jy - 1) = \frac{\pi}{4}$$

$$\arg(-x - 1 + j(1 - y)) = \frac{\pi}{4}$$

$$-(x+1) = -(y-1)$$

$$x + 1 = y - 1 > 0$$

$$x - y = -2.$$

$-(x+1) > 0$ $\quad$ $-(y-1) < 0$

$x + 1 < 0$ $\quad$ $y - 1 > 0$

$x < -1$ $\quad$ $y > 1$

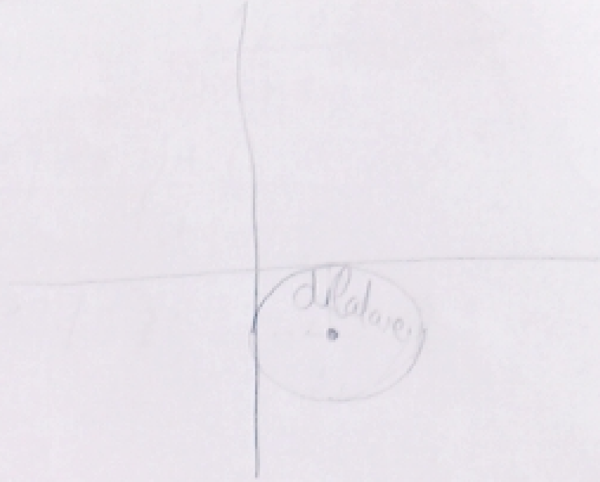
$$|z'| = 1 \Rightarrow \left| \frac{j}{(j - z - 1)^2} \right| = 1$$

$$\frac{1}{|-(z + 1 - j)|} = 1$$

$|z+1-j| = 1$

$z_0 = -1 + j$

$r = 1$



dilatare

$|z'| > 1$

$|z+1-j| < 1$

contractare

7°. $f'(z) = c_2 \left[\frac{-2xy}{(x^2+y^2)^2} + j \frac{y^2-x^2}{(x^2+y^2)^2} \right]$

$\Rightarrow y = 0$ deci $z = x$

$f'(x) = -\frac{c_2 j x^2}{x^4} = -j c_2 \frac{1}{x^2}$

$f(x) = j c_2 \cdot \frac{1}{x} + c_3$

$f(z) = j c_2 \cdot \frac{1}{z} + c_3$

Transformata Fourier integrală

Seminar 9

29 aprilie 2014

$$\mathcal{F}\{f(t)\}(\omega) = \int_{-\infty}^{\infty} e^{-j\omega t} f(t)\, dt \; ; \omega \in \mathbb{R}$$ - spectrul semnalului f.

$$\int_{-\infty}^{\infty} e^{j\omega x} f(x)\, dx = 2\pi j \sum_{\operatorname{Im} z_k > 0} \operatorname{Rez}(f; z_k) + \pi j \sum_{\operatorname{Im} z_k = 0} \operatorname{Rez}(f; z_k)$$

- $\omega > 0$
- $\lim_{|x| \to \infty} f(x) = 0$

$$\begin{cases} \text{I} \quad \omega < 0 \Rightarrow -\omega > 0 \\ \text{II} \quad \omega > 0 \Rightarrow -t = x \end{cases}$$

1. $f(t) = \dfrac{3t-2}{t^2-2t+10}$

Să se calculeze spectrul Fourier - amplitudine și fază - în frecvență

$|\mathcal{F}|$ - amplitudine

$\arg \mathcal{F}$ - fază

$$\mathcal{F}\{f(t)\}(\omega) = \int_{-\infty}^{\infty} e^{j\omega t} \cdot \frac{3t-2}{t^2-2t+10}\, dt \; ; \omega \in \mathbb{R}$$

I $\omega < 0$; $-\omega > 0$

$$\mathcal{F}\{f(t)\}(\omega) = 2\pi j \operatorname{Rez}\left(\frac{e^{-j\omega t}(3t-2)}{t^2-2t+10} ; 1+3j \right)$$

pol simplu

$t^2 - 2t + 1 + 9 = 0$

$(t-1)^2 + 9 = 0$

$\Rightarrow t_1 = 1+3j$

$$= 2\pi j \left. \frac{e^{-j\omega t}(3t-2)}{2t-2} \right|_{1+3j}$$

$$= \pi j \frac{e^{-j\omega(1+3j)}(3+9j-2)}{\cancel{1}+3j-\cancel{1}} = \frac{\pi}{3}$$

II $\omega > 0$; $-t = x$

$\Rightarrow t = -x$

$dt = -dx$

$$\mathcal{F}\{f(t)\}(\omega) = \int_{-\infty}^{\infty} e^{j\omega x} \frac{(-3x-2)}{x^2+2x+10} dx$$

$$= -\int_{-\infty}^{\infty} \frac{e^{j\omega x}(3x+2)}{x^2+2x+10} dx$$

$$= -2\pi j \operatorname{Rez}\left(\frac{e^{j\omega x}(3x+2)}{x^2+2x+10} ; -1+3j\right)$$

pol simplu

$$= -2\pi j \left.\frac{e^{j\omega x}(3x+2)}{2x+2}\right|_{-1+3j}$$

$$= -\pi j \frac{e^{j\omega(-1+3j)} \cdot (-3+9j+2)}{3j}$$

$$= -\frac{\pi}{3} e^{-3\omega} \cdot e^{-j\omega}(-1+9j)$$

$$\mathcal{F}\{f(t)\}(\omega) = \frac{\pi}{3}(1-9j) e^{-3\omega} \cdot e^{-j\omega}$$

$$\mathcal{F}\{f(t)\}(\omega) = \begin{cases} \frac{\pi}{3}(1+9j) e^{3\omega} \cdot e^{-j\omega} \\ \frac{\pi}{3}(1-9j) e^{-3\omega} \cdot e^{-j\omega} & ; \omega > 0. \end{cases}$$

$$|\mathcal{F}| = \begin{cases} \frac{\pi}{3}\sqrt{82} \cdot e^{3\omega} ; & \omega < 0 \\ \frac{\pi}{3}\sqrt{82} \cdot e^{-3\omega} ; & \omega > 0 \end{cases}$$

$$|\mathcal{F}\{f\}| = \frac{\pi}{3}\sqrt{82}\, e^{-3|\omega|}$$

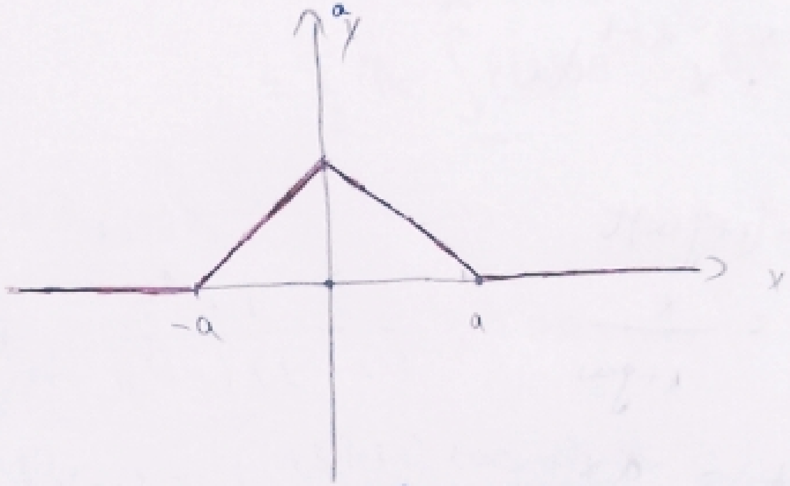
$$\arg \mathcal{F} = \begin{cases} -\omega + \operatorname{arctg} g \ ; & \omega < 0 \\ -\omega + \operatorname{arctg}(-g) \ ; & \omega > 0 \end{cases}$$

Fereastra triunghiulară:

$$l(t) = \begin{cases} a - |t| \ ; & t \leq 0 \\ 0 \ ; & |t| > 0 \end{cases} \qquad a > 0$$

$$\mathcal{F}\{l(t)\}(\omega) = \int_{-a}^{a} (a - |t|) e^{-j\omega t} \, dt$$

$$= \int_{-a}^{a} (a - |t|) \cos \omega t \, dt - j \underbrace{\int_{-a}^{a} (a - |t|) \sin \omega t \, dt}_{=0}$$



$$= 2 \int_{0}^{a} (a - t) \cos \omega t \, dt$$

$u = a - t$

$u' = -1$

$v' = \cos \omega t$

$v = \dfrac{\sin \omega t}{\omega}$

$$= 2 \left[\frac{a-t}{\omega} \sin \omega t \Big|_{0}^{a} + \frac{1}{\omega} \int_{0}^{a} \sin \omega t \, dt \right]$$

$$= 2 \left[-\frac{1}{\omega^2} \cos \omega t \right] \Big|_{0}^{a}$$

$$= \frac{2}{\omega^2} (1 - \cos \omega a)$$

$$\mathcal{F}(l) = \frac{2}{\omega^2} (1 - \cos \omega a) \Rightarrow |\mathcal{F}| = \mathcal{F}$$

$$\arg \mathcal{F} = 0$$

2. $f: \mathbb{R} \to \mathbb{K}$

$$f(t) = t^n \cdot e^{-t}\, u(t) \quad ; n \in \mathbb{N}$$

$$u(t) = \begin{cases} 0 ; t < 0 \\ 1 ; t \geq 0 \end{cases} \quad \Rightarrow \begin{cases} 0 ; t < 0 \\ \frac{1}{2} ; t = 0 \\ 1 ; t > 0 \end{cases}$$

$$\mathcal{F}\{f\}(\omega) = \int_{-\infty}^{\infty} t^n \cdot e^{-t} \cdot u(t) \cdot e^{-j\omega t}\, dt$$

$$= \int_0^{\infty} e^{-(1+j\omega)t} \cdot t^n\, dt$$

$$\Gamma(\alpha) = \int_0^{\infty} e^{-x} \cdot x^{\alpha - 1}\, dx$$

$$x = (1+j\omega)t$$

$$t = \frac{x}{1+j\omega}$$

$$dt = \frac{dx}{1+j\omega}$$

$$\mathcal{F}\{f(t)\}(\omega) = \int_0^{\infty} e^{-x} \left(\frac{x}{1+j\omega}\right)^n \frac{dx}{1+j\omega}$$

$$= \frac{1}{(1+j\omega)^{n+1}} \int_0^{\infty} e^{-x} \cdot x^{(n+1)-1}\, dx$$

$$= \frac{\Gamma(n+1)}{(1+j\omega)^{n+1}}$$

$$\underset{\alpha > 0}{=} \frac{n!\,(1+j\omega)^{n+1}}{(1+\omega^2)^{n+1}}$$

$$\Gamma(n+1) = n!$$

$$|\mathcal{F}| = \frac{n!}{(1+\omega^2)^{\frac{n+1}{2}}}$$

$$\arg \mathcal{F} = \arg \frac{n!}{(1+\omega^2)^{n+1}} + \arg (1-j\omega)^{n+1}$$

$$= (n+1) \arg (1-j\omega)$$

$$= (n+1) (\arg(-\omega) + \varepsilon\pi) \; ; \; (\mathrm{mod}\ 2\pi)$$

$$\mathcal{F}_c \{ f(t) \}(\omega) = \int_0^\infty f(t) \cos \omega t \, dt = \frac{1}{2} \int_{-\infty}^{\infty} f(t) \cos t\omega, \; \omega > 0$$

$$= \frac{1}{2} \operatorname{Re} \int_{-\infty}^{\infty} f(t) e^{j\omega t} dt$$

3°. $f : \mathbb{R}_+ \to \mathbb{K}$

$$f(t) = \frac{t^2+1}{(t^2+4)(t^2+a^2)} \; ; \; a > 0$$

$$\mathcal{F}_c \{ f \}(\omega) = \int_0^\infty \frac{(t^2+1) \cos \omega t}{(t^2+4)(t^2+a^2)} dt$$

$$= \frac{1}{2} \int_{-\infty}^{\infty} \frac{(t^2+1) \cdot \cos \omega t}{(t^2+4)(t^2+a^2)} dt + \frac{1}{2j} \int_{-\infty}^{\infty} \frac{(t^2+1) \sin \omega t}{(t^2+4)(t^2+a^2)} dt$$

$$= \frac{1}{2} \operatorname{Re} \int_{-\infty}^{\infty} \frac{(t^2+1) e^{-j\omega t}}{(t^2+4)(t^2+a^2)} dt$$

$$\int_{-\infty}^{\infty} = \lim_{A \to \infty} \int_{-A}^{A}$$

$$\int_{-\infty}^{\infty} = \lim_{\substack{A \to -\infty \\ B \to \infty}} \int_{A}^{B}$$

$\frac{1}{2}\text{Re}.\ 2\pi j\,(R_{2j} + R_{aj})$ dacă $a \neq 2$

sau

$= \frac{1}{2}\text{Re}\ 2\pi \cdot R_{2j}$ dacă $a = 2$

I $a \neq 2$

$2j$ și aj - poli simpli

$$\text{Rez}(f, z_k) = \frac{(z^2+1)e^{jwt}}{2t(t^2+a^2) + 2t(t^2+1)}$$

$$R_{2j} = \frac{(-4+1)e^{-2w}}{4j(-4+a^2)}$$

$$R_{aj} = \frac{(-a^2+1)e^{-aw}}{2aj(-a^2+4)}$$

$$F_c(f)(w) = \frac{1}{2}\text{Re}\ 2\pi j\left(\frac{-3e^{-2w}}{4j(a^2-4)} + \frac{(a^2-1)e^{-aw}}{2aj(a^2-4)}\right)$$

$$= \frac{\pi}{2}\text{Re}\left(\frac{-3e^{-2w}}{2(a^2-4)} + \frac{(a^2-1)e^{-aw}}{a(a^2-4}\right)$$

$$= \frac{\pi}{2(a^2-4)}\left(\frac{-3e^{-2w}}{2} + \frac{(a^2-1)e^{-aw}}{a}\right)$$

II $a = 2$

$z = 2j$ - pol dublu

z_k - pol de ordin n

$$\text{Rez}\ z_k = \frac{1}{(n-1)!}\lim_{z \to z_k}\left[(z-z_k)^n f(z)\right]^{(n-1)}$$

Seminar 10
6. mai 2014.

1°. Să se rezolve ecuațiile integrale:

a) $\int_0^\infty f(t)\cos\omega t\,dt = \frac{\omega^2}{(\omega^2+1)^3}$; $\omega > 0$

$F_c(\omega) = \int_0^\infty f(t)\cos\omega t\,dt$; $\omega > 0$

$f(x) = \frac{2}{\pi}\int_0^\infty \cos\omega x \int_0^\infty f(t)\cos\omega t\,dt\,d\omega$

$= \frac{2}{\pi}\int_0^\infty F_c(\omega)\cos\omega x\,dx$

$\Rightarrow f(x) = \frac{2}{\pi}\int_0^\infty \frac{\omega^2}{(\omega^2+1)^3}\cos\omega x\,d\omega$

$= \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\omega^2}{(\omega^2+1)^3}\cos\omega x\,d\omega$

$(\omega^2+1)^3 = 0$

$\omega = \pm j$

$= \frac{1}{\pi}\mathrm{Re}\int_{-\infty}^{\infty} \frac{\omega^2 e^{j\omega x}}{(\omega^2+1)^3}\,d\omega$

$= \frac{1}{\pi}\mathrm{Re}\, 2\pi j\, \mathrm{Rez}\left(\frac{\omega^2 e^{j\omega x}}{(\omega^2+1)^3} ; j\right)$

$\mathrm{Rez}\left(\frac{\omega^2 e^{j\omega x}}{(\omega^2+1)^3} ; j\right) = \frac{1}{2!}\lim_{\omega\to j}\left[\frac{\omega^2 e^{j\omega x}}{(\omega+j)^3\cancel{(\omega-j)^3}}\cancel{(\omega-j)^3}\right]''$

$= \frac{1}{2}\lim_{\omega\to j}\left[\frac{(2\omega e^{j\omega x}+\omega^2 jx\cdot e^{j\omega x})(\omega+j)^3}{(\omega+j)^6}\right.$

$\left. - \frac{\omega^2 e^{j\omega x}\cdot 3(\omega+j)^2}{(\omega+j)^6}\right]$

$$= \frac{1}{2}\lim_{\omega\to j}\left[\frac{\omega e^{j\omega x}\left[(2+\omega j x)(\omega+j)-3\omega\right]}{(\omega+j)^4}\right]'$$

$$= \frac{1}{2}\lim_{\omega\to j}\frac{\left\{e^{j\omega x}\left[(2+\omega jx)(\omega+j)-3\omega\right]+\omega jx\, e^{j\omega x}\left[(2+\omega jx)(\omega+j)-3\omega\right]\right.}{(\omega+j)^8}$$

$$\frac{\left.+\omega e^{j\omega x}\left[jx(\omega+j)+(2+\omega jx)-3\right]\right\}(\omega+j)^4-\omega e^{j\omega x}\left[(2+\omega jx)(\omega+j)-3\omega\right]4(\omega+j)^3}{(\omega+j)^8}$$

$$= \frac{1}{2}\lim_{\omega\to j}\frac{e^{j\omega x}\left\{\left[(2+\omega jx)(j+\omega)-3\omega\right](1+\omega jx)-\omega\left[jx(\omega+j)-1+\omega jx\right]\right\}(\omega+j)}{(\omega+j)^5}$$

$$-\frac{4\omega e^{j\omega x}\left[(2+\omega jx)(\omega+j)-3\omega\right]}{(\omega+j)^5}$$

$$= \frac{1}{2}\,\frac{e^{-x}\left\{\left[(2-x)2j-3j\right](1-x)+2xj+j-jx\right\}2j-4je^{-x}\left[(2-x)2j-3j\right]}{(2j)^5}$$

$$= \frac{e^{-x}}{2j}\cdot\left[\left(j-jx-2jx+2jx^2+jx-j\right)2j-4j\cdot\left(4j-2jx-3j\right)\right]$$

$$= \frac{e^{-x}}{2j}\left[\cancel{-2}+x+4x-4x^2-2x\cancel{+2}+4-8x\right]$$

$$= \frac{e^{-x}}{2j}\left(-4x^2-5x+4\right)$$

$$\Rightarrow \frac{1}{\pi}\,\mathrm{Res}\;\cancel{2\pi j}\cdot\frac{e^{-x}}{\cancel{2j}}\left(-4x^2-5x+4\right)=e^{-x}\left(-4x^2-5x+4\right)$$

2°. $\int_0^\infty f(t) \sin \omega t \, dt = e^{-2\omega}$; $\omega > 0$

$$f(x) = \frac{2}{\pi} \int_0^\infty \sin \omega x \underbrace{\int_0^\infty f(t) \sin \omega t \, dt}_{F_s(\omega)} \, d\omega$$

$$f(x) = \frac{2}{\pi} \int_0^\infty F_s(\omega) \sin \omega x \, d\omega$$

deci: $f(x) = \frac{2}{\pi} \int_0^\infty e^{-2\omega} \sin \omega x \, d\omega$

$$= \frac{\cancel{2}}{\pi} \int_0^\infty e^{2\omega} \cdot \frac{e^{j\omega x} - e^{-j\omega x}}{\cancel{2}j} \, d\omega$$

$$= \frac{1}{\pi} \int_0^\infty \frac{e^{\omega(2+jx)} - e^{\omega(2-jx)}}{j} \, d\omega$$

$$= \frac{1}{\pi j} \left[\frac{e^{(-2+jx)\omega}}{-2+jx} - \frac{e^{-(2+jx)\omega}}{-(2+jx)} \right] \Bigg|_0^\infty$$

$$= \frac{-1}{\pi j} \left(\frac{1}{-2+jx} + \frac{1}{2+jx} \right)$$

$$= \frac{1}{\pi j} \left(\frac{1}{2-jx} - \frac{1}{2+jx} \right)$$

$$= \frac{1}{\pi j} \cdot \frac{\cancel{2} + jx \cancel{-2} + jx}{4 + x^2}$$

$$= \frac{2x}{\pi (4 + x^2)}$$

$$X(m) = \sum_{n=0}^{N-1} x(n) w^{-mn} \quad ; \forall m \in \mathbb{Z}$$

$$x(n+kN) = x(n)$$

$$X(m+kN) = X(m)$$

$$w = e^{\frac{2\pi j}{N}}$$

Exemple.

1°. $x(n) = 1 \,;\, x \in K^{100}$

$$w = e^{\frac{2\pi j}{100}} = e^{\frac{\pi j}{50}}$$

$$X(m) = \sum_{n=0}^{99} 1 \cdot e^{-\frac{j\pi}{50} mn} = \sum_{n=0}^{99} e^{-\frac{\pi j}{50} mn}$$

$$= \sum_{n=0}^{99} \left(e^{-\frac{j\pi m}{50}} \right)^n$$

$$1 + q + \dots + q^{99} = \frac{q^{99} \cdot q - 1}{q - 1} \quad ; \; q \neq 1$$

$$\Rightarrow X(m) = \begin{cases} \dfrac{e^{-\frac{j\pi m}{50} \cdot 100} - 1}{e^{-\frac{j\pi m}{50}} - 1} \;;\; e^{-\frac{j\pi m}{50}} \neq 1 \;;\; m \not\equiv 0 \pmod{100} \\ 100 \;;\; e^{-\frac{j\pi m}{50}} = 1 \\ \quad m \equiv 0 \pmod{100} \end{cases}$$

2°. $x(n) = \left(\frac{j-1}{\sqrt{2}} \right)^n \,;\, x \in K^{50} \quad \Rightarrow N = 50$

$$w = e^{\frac{2\pi j}{50}} = e^{\frac{\pi j}{25}}$$

$$X(m) = \sum_{n=0}^{49} \left(\frac{j-1}{\sqrt{2}} \right)^n \cdot e^{-j\frac{\pi}{25} m \cdot n}$$

$$\left|\frac{j-1}{\sqrt{2}}\right| = \frac{\sqrt{2}}{\sqrt{2}} = 1$$

$$\Rightarrow \arg\left(\frac{j-1}{\sqrt{2}}\right) = \frac{3\pi}{4}$$

Vậy: $\frac{j-1}{\sqrt{2}} = e^{j\frac{3\pi}{4}}$

$$\Rightarrow X(m) = \sum_{n=0}^{49} e^{j\frac{3\pi n}{4}} \cdot e^{-j\frac{mn\pi}{25}}$$

$$= \sum_{n=0}^{49} e^{j\left(\frac{3\pi}{4} - \frac{m\pi}{25}\right)n}$$

$$= \sum_{n=0}^{49} \left(e^{j\pi\frac{75-4m}{100}}\right)^n$$

$$= \frac{e^{j\pi\frac{75-4m}{100}\cdot 50} - 1}{e^{j\pi\cdot\frac{75-4m}{100}} - 1}$$

$$= \frac{e^{j\pi\frac{(75-4m)}{2}} - 1}{e^{j\pi\frac{(75-4m)}{100}} - 1}$$

$$= \frac{e^{j\pi\frac{75}{2}} \cdot e^{-2mj\pi} = 1 \;- 1}{e^{j\pi\frac{(75-4m)}{100}}}$$

$$= \frac{e^{j\pi\frac{3}{2}} - 1}{e^{j\pi\frac{(75-4m)}{100}}}$$

$$= \frac{-j-1}{e^{j\pi\frac{(75-4m)}{100}} - 1} \; ; m \in \mathbb{Z}$$

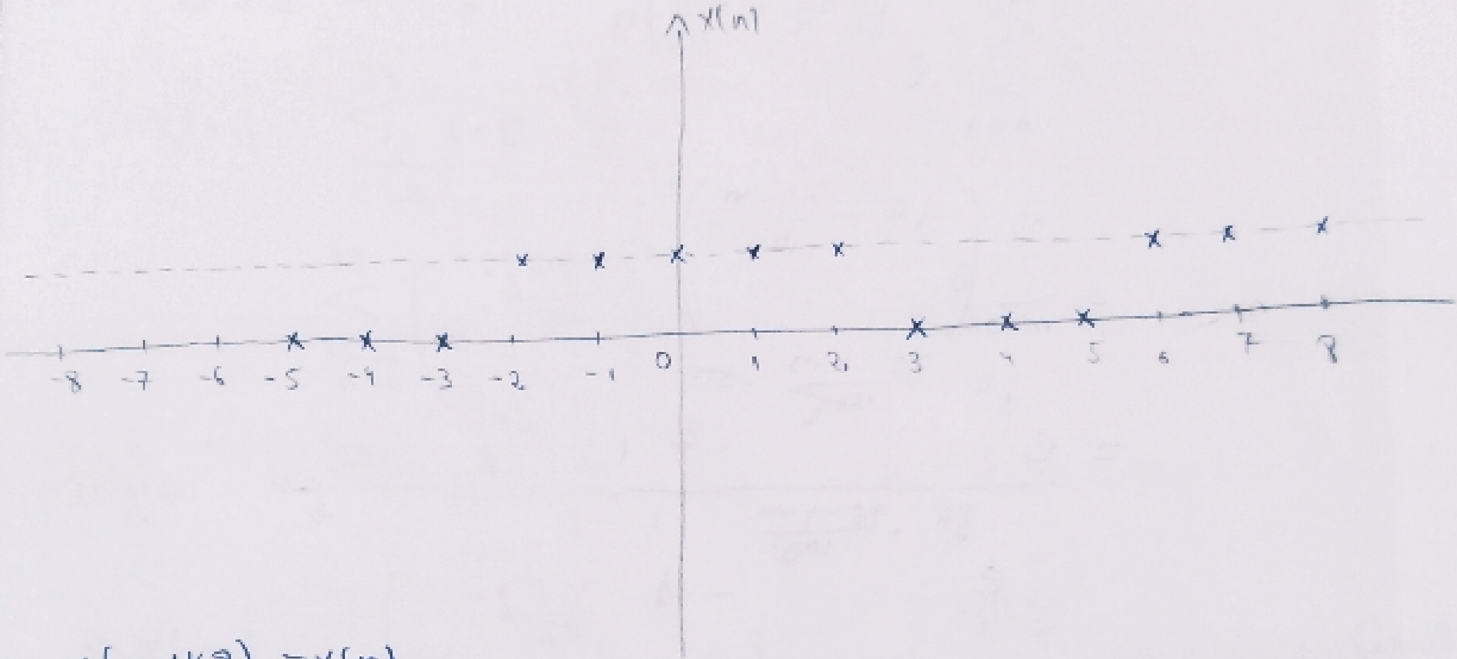
3. $x \in h$

$$x(n) = \begin{cases} 1; & n \in \{0;1;2;6,7\} \\ 0; & n \in \{3,4,5\} \end{cases}$$

$$\mathcal{F}_d\{x;n\}(m) = ?$$

$$w = e^{\frac{2\pi j}{8}} = e^{\frac{\pi j}{4}}$$

$$\sum_{m=1}^{7} \frac{1-\cos 5\frac{m\pi}{4}}{1-\cos\frac{m\pi}{4}} = 15$$



$$x(n+k8) = x(n)$$

$$X(m) = \sum_{n=0}^{7} x(n)\, e^{-\frac{\pi j m n}{4}}$$

$$= \sum_{n=-2}^{2} 1 \cdot e^{-\frac{\pi j m n}{4}} + \sum_{n=3}^{5} 0 \cdot e^{-\frac{\pi j m n}{4}}$$

$$= e^{\frac{\pi j m}{2}} + e^{\frac{\pi j m}{4}} + 1 + e^{-\frac{\pi j m}{4}} + e^{-\frac{\pi j m}{2}}$$

$$= e^{\frac{\pi j m}{2}}\left(1 + e^{-\frac{m\pi j}{4}} + e^{-\frac{2m\pi j}{4}} + e^{-\frac{3m\pi j}{4}} + e^{-\frac{4m\pi j}{4}}\right)$$

$$= e^{\frac{m j \pi}{2}} \frac{e^{-\frac{5 m \pi j}{4}} - 1}{e^{-\frac{m \pi j}{4}} - 1} = e^{\frac{m j \pi}{2}} \cdot \frac{e^{-\frac{5 m \pi j}{8}}\left(e^{-\frac{5 m \pi j}{8}} - e^{\frac{5 m \pi j}{8}}\right)}{e^{-\frac{m \pi j}{8}}\left(e^{-\frac{m \pi j}{8}} - e^{\frac{m \pi j}{8}}\right)}$$

$m \neq 0$

pt $m = 0 \Rightarrow X(0) = 5$

$$= e^{\frac{m \pi j}{2} - \frac{5 m \pi j}{8} + \frac{m \pi j}{8}} \frac{\cancel{-2j} \sin \frac{5 m \pi}{8}}{\cancel{-2j} \sin \frac{m \pi}{8}}$$

! Parseval !

$$= \frac{\sin \frac{5 m \pi}{8}}{\sin \frac{m \pi}{8}}$$

$$X(m) = \begin{cases} \dfrac{\sin \frac{5 m \pi}{8}}{\sin \frac{m \pi}{8}}; & m \neq 0 \\ 5; & m = 0 \end{cases}$$

$$\boxed{\sum_{m=0}^{N-1} |X(m)|^2 = N \sum_{n=0}^{N-1} |x(n)|^2}$$

deci: $$\sum_{m=0}^{7} |X(m)|^2 = 8 \sum_{n=0}^{7} |x(n)|^2 = 8 \cdot 5 = 40$$

$$|X(0)|^2 + \sum_{m=1}^{7} |X(m)|^2 = 40$$

$$\sum_{m=1}^{7} \left| \frac{\sin \frac{5 m \pi}{8}}{\sin \frac{m \pi}{8}} \right|^2 = 40 - 25$$

$$\sum_{m=1}^{7} \frac{\sin^2 \frac{5 m \pi}{8}}{\sin^2 \frac{m \pi}{8}} = 15$$

$$\sum_{m=1}^{7} \frac{1 - \cos \frac{5 m \pi}{4}}{1 - \cos \frac{m \pi}{4}} = 15$$

i.

$$x(n) = \begin{pmatrix} j-2 \\ 0 \\ 1 \\ 2j \end{pmatrix}$$

$$X(m) = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{pmatrix} \begin{pmatrix} j-2 \\ 0 \\ 1 \\ 2j \end{pmatrix}$$

$$= \begin{pmatrix} j-2+1+2j \\ j-2+1-2 \\ j-2+1-2j \\ j-\cancel{2}-1+\cancel{2} \end{pmatrix} = \begin{pmatrix} 3j-1 \\ j-5 \\ -j-1 \\ j-1 \end{pmatrix}$$

$$1+9+25+1+1+1+1+1 = 4(1+4+1+4)$$

$$40 = 4 \cdot 10 \text{ — „A"}$$

Seminar 11

13 mai 2014

Transformata Laplace

1°. $x(n) = ?$

$$X(m) = (3+j;\ 2+2j;\ -1+j;\ -2j)^T$$

$$X(m) = W \cdot x(n)$$

$$x = \frac{1}{N} \cdot \overline{W} X$$

$$W = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{pmatrix} \Rightarrow \overline{W} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{pmatrix}$$

$$x = \overline{W} X = \frac{1}{4} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{pmatrix} \begin{pmatrix} 3+j \\ 2+2j \\ -1+j \\ -2j \end{pmatrix}$$

$$= \frac{1}{4} \begin{pmatrix} 3+\cancel{j}+2+\cancel{2j}-1-\cancel{j}-\cancel{2j} \\ 3+j+\cancel{2j}-2-\cancel{2j}\cancel{-2}-2j+\cancel{2} \\ 3+\cancel{j}-2-\cancel{2j}-1-\cancel{j}+\cancel{2j} \\ 3+\cancel{j}-\cancel{2j}+2+1+\cancel{j}+2 \end{pmatrix}$$

$$= \frac{1}{4} \begin{pmatrix} 4 \\ 1-j \\ 0 \\ 8 \end{pmatrix} = \begin{pmatrix} 1 \\ \frac{1-j}{4} \\ 0 \\ 2 \end{pmatrix}$$

f	F
$u(t)=\begin{cases}0 ; t<0\\ 1 ; t>0\end{cases}$	$\frac{1}{s}$
$t^n ; n\in\mathbb{N}$	$\frac{n!}{s^{n+1}}$
t^α	$\frac{\Gamma(\alpha+1)}{s^{\alpha+1}} ; \alpha>-1$
e^{-at}	$\frac{1}{s+a}$
$\cos\omega t$	$\frac{s}{s^2+\omega^2}$
$\sin\omega t$	$\frac{\omega}{s^2+\omega^2}$
$\operatorname{ch}\omega t$	$\frac{s}{s^2-\omega^2}$
$\operatorname{sh}\omega t$	$\frac{\omega}{s^2-\omega^2}$

$$\Gamma(\alpha)=\int_0^\infty e^{-x}\cdot x^{\alpha-1}\,dx \qquad \alpha>0$$

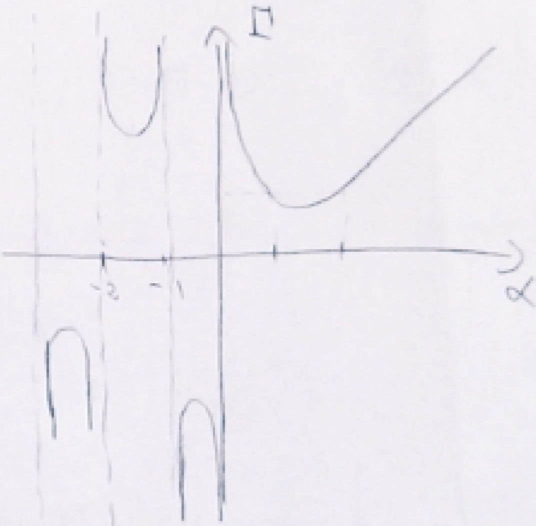
$$\Gamma(\alpha+1)=\alpha\Gamma(\alpha)$$

$$\Gamma(\alpha+n+1)=\alpha(\alpha+1)\dots(\alpha+n)\Gamma(\alpha)$$

$$\alpha=n\in\mathbb{N} \Rightarrow \Gamma(n+1)=n!$$

$$\Gamma(1)=1$$

$$\Gamma(2)=1$$



2°. Să se calculeze următoarele transformate Laplace.

$$\boxed{F(s)=\mathcal{L}\{f(t)\}(s)=\int_0^\infty e^{-st}\cdot f(t)\cdot dt}$$

a). $f(t)=e^{-5t}\cos^2 3t$

$$=e^{-5t}\cdot\frac{1+\cos 6t}{2}$$

$$=\frac{1}{2}e^{-5t}+\frac{1}{2}e^{-5t}\cdot\cos 6t.$$

$$\mathcal{L}\{e^{-at}\cdot f(t)\}(s)=F(s+a)=\mathcal{L}\{f(t)\}(s+a)$$

$$F(s)=\frac{1}{2}\cdot\frac{1}{s+5}+\frac{1}{2}\cdot\frac{s+5}{(s+5)^2+36}$$

b) $$f(t)=\frac{3e^{2t}-5te^{-10t}}{\sqrt{t}}$$

$$=3t^{-\frac{1}{2}}\cdot e^{2t}-5\cdot t^{\frac{1}{2}}\cdot e^{-10t}$$

$$\mathcal{L}\{t^{-\frac{1}{2}}\}(s)=\frac{\Gamma(-\frac{1}{2}+1)}{s^{-\frac{1}{2}+1}}=\frac{\Gamma(\frac{1}{2})}{\sqrt{s}}=\frac{\sqrt{\pi}}{\sqrt{s}}=\sqrt{\frac{\pi}{s}}$$

$$\mathcal{L}\{t^{\frac{1}{2}}\}(s)=\frac{\Gamma(\frac{1}{2}+1)}{s^{\frac{1}{2}+1}}=\frac{\frac{1}{2}\cdot\Gamma\frac{1}{2}}{s\sqrt{s}}=\frac{1}{2s}\sqrt{\frac{\pi}{s}}$$

$$\Rightarrow F(s)=3\sqrt{\frac{\pi}{s-2}}+\frac{1}{2(s+10)}\sqrt{\frac{\pi}{s+10}}$$

c) $$f(t)=\frac{\sin at+\cos bt-1}{t};\ a,b\neq 0.$$

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\}(s)=\int_s^\infty \mathcal{L}\{f(t)\}(u)\,du$$

$$F(s)=\int_s^\infty \mathcal{L}\{\sin at+\cos bt-1\}(u)\,du$$

$$=\int_s^\infty\left(\frac{a}{u^2+a^2}+\frac{u}{u^2+b^2}-\frac{1}{u}\right)du.$$

$$=\left(\operatorname{arctg}\frac{u}{a}+\frac{1}{2}\ln(u^2+b^2)-\ln u\right)\Big|_s^\infty$$

$$=\operatorname{arctg}\frac{u}{a}\Big|_s^\infty+\ln\frac{\sqrt{u^2+b^2}}{u}\Big|_s^\infty$$

$$=\frac{\pi}{2}-\operatorname{arctg}\frac{s}{a}-\ln\frac{\sqrt{s^2+b^2}}{s}$$

$= \operatorname{arctg} \frac{a}{s} + \ln \frac{s}{\sqrt{s^2+b^2}}$

Temă:

$$f(t) = \frac{t^2 \operatorname{sh} t + 2 \operatorname{ch} 3t}{t}$$

Polinoamele LAGUERRE:

$$L_n(t) = \frac{1}{n!} \cdot e^t (t^n e^{-t})^{(n)} \ ; n \geq 0$$

$$\mathcal{L}\{f'(t)\}(s) = sF(s) - f(+0)$$

$$\mathcal{L}\{f^{(n)}(t)\}(s) = s^n F(s) - s^{n-1} f(+0) - s^{n-2} f'(+0) - \dots - f^{(n-1)}(+0)$$

$$\mathcal{L}\{t^n e^{-t}\}(s) = \frac{n!}{(s+1)^{n+1}}$$

$$\mathcal{L}\{(t^n e^{-t})^{(n)}\}(s) = s^n \cdot \frac{n!}{(s+1)^{n+1}} - \underbrace{s^{n-1} \cdot (t^n e^{-t})\Big|_{t=+0}}_{=0} - \underbrace{s^{n-2} (t^n e^{-t})'\Big|_{t=+0}}_{=0}$$

$$- \underbrace{s^{n-3} (t^n e^{-t})''\Big|_{t=+0}}_{=0} - \dots - \underbrace{(t^n e^{-t})^{(n-1)}\Big|_{t=+0}}_{?}$$

$$(t^n e^{-t})^{(k)} = \sum_{i=0}^{k} C_k^i (t^n)^{(i)} \cdot (e^{-t})^{(k-i)}$$

$$= t^n (e^{-t})^{(k)} - C_k^1 (t^n)' (e^{-t})^{(k-1)} - \dots$$

$$\Rightarrow \mathcal{L}\{(t^n e^{-t})^{(n)}\}(s) = s^n \cdot \frac{n!}{(s+1)^{n+1}}$$

$$\mathcal{L}\{e^t (t^n e^{-t})^{(n)}\}(s) = (s-1)^n \cdot \frac{n!}{(s-1+1)^{n+1}}$$

$$\boxed{\mathcal{L}\{L_n(t)\}(s) = \frac{(s-1)^n}{s^{n+1}}}$$

$$Si\,t = \int_0^\infty \frac{\sin u}{u}\, du \qquad \mathcal{L}\left\{\int_0^t f(u)\,du\right\}(s) = \frac{1}{s}\mathcal{L}\{f(t)\}(s)$$

sinus integral

$$\mathcal{L}\{Si\,t\}(s) = \frac{1}{s}\mathcal{L}\left\{\frac{\sin t}{t}\right\}(s)$$

$$= \frac{1}{s}\int_s^\infty \mathcal{L}\{\sin t\}(y)\, dy$$

$$= \frac{1}{s}\int_s^\infty \frac{1}{y^2+1}\, dy$$

$$= \frac{1}{s} \operatorname{arctg} y \Big|_s^\infty$$

$$= \frac{1}{s}\left(\frac{\pi}{2} - \operatorname{arctg} s\right)$$

$$= \frac{\pi}{s} \operatorname{arctg} \frac{1}{s}$$

$$Ci(t) = \int_t^\infty \frac{\cos x}{x}\,dx \,;\, t>0$$

cosinus integral

$x = t \cdot y$

$dx = t\,dy$

y - noua variabilă

pt $x = t \Rightarrow y = 1$

$x = \infty \Rightarrow y = \infty$

$$\Rightarrow Ci(t) = \int_1^\infty \frac{\cos ty}{ty} \cdot t\,dy$$

$$= \int_1^\infty \frac{\cos ty}{y}\,dy$$

$$\mathcal{L}\{f(t,x)\}(s) = F(s,x)$$

$$\mathcal{L}\{Ci(t)\}(s) = \mathcal{L}\left\{\frac{\cos ty}{y}\,dy\right\}(s) =$$

$$= \int_1^\infty \mathcal{L}\left\{\frac{\cos ty}{y}\right\}(s)\,dy = \int_1^\infty \frac{1}{y}\,\mathcal{L}\{\cos ty\}(s)\,dy$$

$$= \int_1^\infty \frac{1}{y} \cdot \frac{s}{s^2+y^2}\,dy$$

$$= s\int_1^\infty \frac{y\,dy}{y^2(s^2+y^2)} = \frac{1}{s}\int_1^\infty \left(\frac{y}{y^2} - \frac{y}{y^2+s^2}\right)dy$$

$$= \frac{1}{s}\left[\ln y\Big|_1^\infty - \frac{1}{2}\ln(y^2+s^2)\Big|_1^\infty\right] = \frac{1}{s}\ln\frac{y}{\sqrt{y^2+s^2}}\Big|_1^\infty = \frac{1}{s}\ln\sqrt{s^2+1}$$

3°. a) $f(t) = \sin\sqrt{t}$

$$\begin{tikzcd} \sin\sqrt{t} \arrow[r, "\sqrt{t}=\tau"] & \sin\tau \arrow[d] \\ {} \arrow[r, "?"] & \frac{1}{\tau^2+1} \end{tikzcd}$$

$$\Rightarrow f(t) = \sin\sqrt{t} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \cdot (\sqrt{t})^{2n+1}$$

$$\left.\begin{array}{l} \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} \cdot x^{2n} \\ e^x = \sum_{n=0}^{\infty} \frac{1}{n!} \cdot x^n \end{array}\right\} x \in \mathbb{R}$$

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n ; \ |x| < 1$$

$$\Rightarrow \sin\sqrt{t} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \cdot t^{n+\frac{1}{2}}$$

$$F(s) = \mathcal{L}\left\{ \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \cdot t^{n+\frac{1}{2}} \right\}(s)$$

Reciproca T. lui Heaviside $= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \mathcal{L}\left\{ t^{n+\frac{1}{2}} \right\}(s)$

$$\mathcal{L}\left\{ t^{n+\frac{1}{2}} \right\}(s) = \frac{\Gamma(n+\frac{1}{2}+1)}{s^{n+\frac{1}{2}+1}} = \frac{1}{2}\left(\frac{1}{2}+1\right)\left(\frac{1}{2}+2\right) \cdot \ldots \cdot \left(\frac{1}{2}+n\right)\Gamma\left(\frac{1}{2}\right)$$

$$= \frac{1 \cdot 3 \cdot 5 \cdot \ldots \cdot (2n+1)}{2^{n+1}} \cdot \sqrt{\pi}$$

$$= \frac{(2n+1)!}{2^{2n+1} \cdot n!} \cdot \sqrt{\pi} \qquad (2n)!! = 2^n \cdot n!$$

$$\Rightarrow \mathcal{L}\left\{ t^{n+\frac{1}{2}} \right\}(s) = \frac{(2n+1)! \cdot \sqrt{\pi}}{2^{2n+1} \cdot n! \cdot s^{n+\frac{1}{2}+1}}$$

$$\Rightarrow$$

$$F(s) = \sum_{n=0}^{\infty} \frac{(-1)^n}{\cancel{(2n+1)!}} \cdot \frac{\cancel{(2n+1)!}\sqrt{\pi}}{2^{2n+1} \cdot n! \cdot s^{n+\frac{1}{2}+1}}$$

$$= \sqrt{\pi} \cdot \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \left(\frac{1}{4s}\right)^n \cdot \frac{1}{2s\sqrt{s}}$$

$$= \frac{\sqrt{\pi}}{\sqrt{s}} \cdot \frac{1}{2s} \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \cdot \left(\frac{1}{4s}\right)^n$$

$$= \frac{1}{2s}\sqrt{\frac{\pi}{s}} \cdot e^{-\frac{1}{4s}}$$

Seminar 12
21 mai 2014
M. S.

Transformarea Laplace

<u>Ex 1.</u> $i = \int_0^\infty \underbrace{t\cos^2(2t)}_{f(t)} \underbrace{e^{-3t}}_{s=3}\, dt$

<u>Rezolvare</u>

$$f \in \mathcal{O} \Rightarrow \mathcal{L}\{f(t)\}(s) = \int_0^\infty f(t)\, e^{-st}\, dt$$

$$\mathcal{L}\{e^{at}\}(s) = \frac{1}{s-a} \Leftrightarrow \mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\}(t) = e^{at}u(t)$$

$$\Rightarrow i = \mathcal{L}\{t\cos^2(2t)\}(3) \quad (1)$$

$$\mathcal{L}\{t\cos^2(2t)\}(s) \overset{n=1}{=} (-1)^1 \cdot \left[\mathcal{L}\{\cos^2(2t)\}(s)\right]'$$

$$= -\left(\mathcal{L}\left\{\frac{1}{2}(1+\cos 4t)\right\}(s)\right)'$$

$$= -\frac{1}{2}\left[\mathcal{L}\{1\}(s) + \mathcal{L}\{\cos 4t\}(s)\right]'$$

$$= -\frac{1}{2}\left(\frac{1}{s} + \frac{s}{s^2+16}\right)'$$

$$\mathcal{L}\{t^n f(t)\}(s) = (-1)^n \left(\mathcal{L}\{f(t)\}(s)\right)^{(n)}$$

$$= -\frac{1}{2}\left(-\frac{1}{s^2} + \frac{16-s^2}{(s^2+16)^2}\right) \quad (2)$$

Din (1) și (2) $\Rightarrow i = -\frac{1}{2}\left(-\frac{1}{9} + \frac{7}{(25)^2}\right)$

$$= -\frac{1}{2}\left(\frac{-625+63}{9\cdot 625}\right) = \frac{281}{5625}$$

Ex2 $i = \int_0^\infty \frac{3t^3\sqrt{t}e^{2t}+2\sin 5t}{t}\, e^{-4t}\, dt$

Rezolvare

$$i = \int_0^\infty \left[\frac{3t^3\sqrt{t}\, e^{2t}}{t} + \frac{2\sin 5t}{t}\right] e^{-4t}\, dt$$

$$= 3\int_0^\infty \left(t^{\frac{5}{2}} e^{2t}\cdot e^{-4t}\right) dt + 2\int_0^\infty \frac{\sin 5t}{t}\, e^{-4t}\, dt.$$

$$= 3\, i_1 + 2\, i_2$$

$$i_1 = \int_0^\infty \underbrace{t^{\frac{5}{2}}}_{f(t)} e^{-2t}\, dt = \mathcal{L}\{t^2\sqrt{t}\}(2) = \mathcal{L}\{t^{\frac{5}{2}}\}(+2)$$

$s = +2$

$$= \frac{\Gamma\left(\frac{5}{2}+1\right)}{s^{\frac{5}{2}+1}} = \frac{\frac{5}{2}\Gamma\left(\frac{5}{2}\right)}{2^{\frac{5}{2}+1}}$$

$s = 2$

$$= \frac{5}{2} \cdot \frac{\frac{3}{2}\Gamma\left(\frac{3}{2}\right)}{8\sqrt{2}}$$

$$= \frac{15}{32\sqrt{2}} \cdot \Gamma\left(\frac{3}{2}\right)$$

$$= \frac{15}{32\sqrt{2}} \cdot \frac{1}{2}\Gamma\left(\frac{1}{2}\right) = \frac{15}{64\sqrt{2}} \cdot \sqrt{\pi}$$

$$i_2 = \int_0^\infty \frac{\sin 5t}{t}\, e^{-4t}\, dt = \mathcal{L}\left\{\frac{\sin 5t}{t}\right\}(4)$$

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\}(s) = \int_s^\infty \mathcal{L}\{f(t)\}(y)\, dy$$

$= \int_4^\infty \mathcal{L}\{\sin 5t\}(s)\, ds$

$= \int_4^\infty \frac{5}{s^2+25}\, ds$

$\operatorname{arctg} x + \operatorname{arctg}\frac{1}{x} = \frac{\pi}{2}$

$\frac{\pi}{2} - \operatorname{arctg} x = \operatorname{arctg}\frac{1}{x}$

$= \cancel{5} \cdot \frac{1}{\cancel{5}} \cdot \operatorname{arctg} \frac{s}{5}\Big|_4^\infty$

$= \frac{\pi}{2} - \operatorname{arctg}\frac{4}{5} = \operatorname{arctg}\frac{5}{4}$

$\Rightarrow i = \frac{45\sqrt{\pi}}{64\sqrt{2}} + 2\operatorname{arctg}\frac{5}{4}$

Ex 3 $\quad x''' + 9x' = e^t \; ; \; x(0) = 0$

$x'(0) = 1$

$x''(0) = 2 \; ; \; x = x(t) \in \mathcal{O}$

$\mathcal{L}\{x'''(t)\}(s) + 9\mathcal{L}\{x'(t)\}(s) = \mathcal{L}\{e^t\}(s)$

$X(s) = \mathcal{L}\{x(t)\}(s) \longleftrightarrow x(t) = \mathcal{L}^{-1}\{X(s)\}(t) \; (1)$

$\mathcal{L}\{x'(t)\}(s) = sX(s) - x(0)$

$\mathcal{L}\{x''(t)\}(s) = s^2X(s) - sx(0) - x'(0)$

$\mathcal{L}\{x'''(t)\}(s) = s^3X(s) - s^2(x(0)) - sx'(0) - x''(0)$

$s^3X(s) - s^2 \cdot 0 - s \cdot 1 - 2 + 9(sX(s) - 0) = \frac{1}{s-1}$

$X(s)(s^3 + 9s) = \frac{1}{s-1} + s + 2$

$$X(s) = \frac{1+s^2+s-2}{(s-1)(s^3+9s)}$$

$$= \frac{(s^2+s-1)}{s(s-1)(s^2+9)} = \frac{A}{s} + \frac{B}{s-1} + \frac{Cs+D}{s^2+9}$$

$$\Rightarrow s^2+s-1 = (s-1)(s^2+9)A + (s-1)(s^2+9)B + s(s-1)(Cs+D)$$

pt $s=0 \Rightarrow -1 = -9a \Rightarrow a = \frac{1}{9}$

$s=1 \Rightarrow 1 = 10b \Rightarrow b = \frac{1}{10}$

$s = 3j \Rightarrow -9+3j-1 = 3cj + 1(9-9)$

$-10+3j = -9D+9C+j(-3D-27C)$

$-9D+9C = -10$

$-3D-27C = 3$

$90C = -\frac{19}{1} \Rightarrow C = -\frac{19}{20}$

$\Rightarrow C = -\frac{19}{90}$

$D = \frac{9}{10}$

$$\Rightarrow X(s) = \frac{1}{9}\cdot\frac{1}{s} + \frac{1}{10}\cdot\frac{1}{s-1} - \frac{19}{90}\,\frac{s}{s^2+9} + \frac{9}{10}\cdot\frac{1}{s^2+9}$$

$$\mathcal{L}^{-1}\{X(s)\}(t) = \frac{1}{9}\mathcal{L}^{-1}\left\{\frac{1}{s}\right\}(t) + \frac{1}{10}\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\}(t)$$

$$+ \frac{19}{90}\mathcal{L}^{-1}\left\{\frac{s}{s^2+9}\right\}(t) + \frac{9}{10}\mathcal{L}^{-1}\left\{\frac{1}{s^2+9}\right\}(t)$$

$x(t) = \frac{1}{9}\cdot 1 + \frac{1}{10}\cdot e^{t} - \frac{19}{90}\cos 3t + \frac{9}{10}\cdot\frac{1}{3}\sin 3t]\, u(t)$

verificare: $x(0)=0$.

Ex 4 $\quad x''' + 3x'' - 4x' - 12x = 1$ "Dyu – Ce fac?" "Ahu – Zambește :o3"

$x(0)=0$

$x'(0)=0$

$x''(0)=-1$

$\mathcal{L}\{x'''(t)\}(s) + 3\mathcal{L}\{x''(t)\}(s) - 4\mathcal{L}\{x'(t)\}(s) - 12\mathcal{L}\{x(t)\}(s) = \mathcal{L}\{1\}$

$\mathcal{L}\{x(t)\}(s) = X(s)$

$\Rightarrow s^3X(s) - s^2\cdot x(0) - s\cdot x'(0) - x''(0) + 3[s^2X(s) - s\cdot x(0) - x'(0)]$

$-4[sX(s) - x(0)] - 12\cdot X(s) = \frac{1}{s}$

$X(s)(s^3 + 3s^2 - 4s - 12) = \frac{1}{s} - \frac{s}{s}$

$X(s) = \frac{1-s}{s(s^3+3s^2-4s-12)}$

$= \frac{(1-s)}{s(s+3)(s+2)(s-2)} = \frac{A}{s} + \frac{B}{s+3} + \frac{C}{s+2} + \frac{D}{s-2}$

$(1-s) = A(s+3)(s+2)(s-2) + Bs(s+2)(s-2) + Cs(s+3)(s-2) + Ds(s+3)(s+2)$

pt $s=0$: $1 = 3A\cdot -4 \Rightarrow A = -\frac{1}{12}$

$s=-2$: $\Rightarrow 3 = -2C\cdot 1\cdot(-4)$, $C = \frac{3}{8}$.

$s=2$: $-1 = 2D\cdot 5\cdot 4 \Rightarrow D = -\frac{1}{40}$

$s=-3$: $\Rightarrow 4 = -3B(-1)\cdot(-5) \Rightarrow B = -\frac{4}{15}$

$$X(s) = -\frac{1}{12}\cdot\frac{1}{s} + \left(-\frac{4}{15}\right)\frac{1}{s+3} + \frac{3}{8}\cdot\frac{1}{s+2} - \frac{1}{40}\cdot\frac{1}{s-2} \quad \Big| \mathcal{L}^{-1}$$

$$x(t) = -\frac{1}{12}\cdot 1 - \frac{4}{15}\cdot e^{-3t} + \frac{3}{8}e^{-2t} - \frac{1}{40}e^{2t}$$

verificare $x(0)=0$

$$-\frac{1}{12} - \frac{4}{15} + \frac{3}{8} - \frac{1}{40} = \frac{14}{40} + \left(-\frac{5+16}{60}\right)$$

$$= \frac{14}{40} - \frac{11}{60} = \frac{840-440}{240} =$$

Ex 5 $\quad x'' + 9x = \frac{1}{\cos 3t} \quad ; x(0)=3$

$\qquad x'(0)=6.$

$$\mathcal{L}\{x''(t)\}(s) + 9\,\mathcal{L}\{x(t)\}(s) = \mathcal{L}\underbrace{\left\{\frac{1}{\cos 3t}\right\}}(s)$$

$$\mathcal{L}\{x(t)\}(s) = X(s)$$

$$s^2\cdot X(s) - s\cdot x(0) - x'(0) + 9X(s) = F(s)$$

(1) $F(s) = \mathcal{L}\left\{\frac{1}{\cos 3t}\right\}(s) \iff \mathcal{L}^{-1}\{F(s)\} = \frac{1}{\cos 3t}$

$$(s^2+9)\,X(s) = 3s + 6 + F(s)$$

$$X(s) = \frac{3s}{s^2+9} + \frac{6}{s^2+9} + \frac{F(s)}{s^2+9} \quad \Big| \mathcal{L}^{-1}$$

(2) $x(t) = 3\cdot\cos 3t + \cancel{6}^{2}\cdot\frac{1}{\cancel{3}}\sin 3t + \frac{\mathcal{L}^{-1}\{F(s)\}}{s^2+9}$

$$\mathcal{L}^{-1}\left\{F(s)\cdot\frac{1}{s^2+9}\right\}(s) = \mathcal{L}^{-1}\{F(s)\} * \mathcal{L}^{-1}\left\{\frac{1}{s^2+9}\right\}$$

(1) $\frac{1}{\cos 3t} * \frac{1}{3}\sin 3t$

$$(f*g)(t)=\int_0^t f(x)\cdot g(t-x)\,dx$$

$$\Rightarrow \frac{1}{3}\int_0^t \frac{1}{\cos 3x}\cdot \sin 3(t-x)\,dx$$

$$=\frac{1}{3}\int_0^t \frac{\sin 3t\cdot\cos 3x-\sin 3x\cdot\cos 3t}{\cos 3x}\,dx$$

$$=\frac{1}{3}\left[\int_0^t \sin 3t\,dx-\int_0^t \cos 3t\,(tg\,3x)\,dx\right]$$

$$=\frac{1}{3}\left[t\sin 3t-\cos 3t\cdot\frac{1}{3}\left(\ln|\cos 3x|\right)\Big|_0^t\right]$$

$$=\frac{1}{3}\left[t\sin 3t-\cos 3t\cdot\frac{1}{3}\ln\cos 3t\right] \quad (3)$$

din (2) şi (3)

$$\Rightarrow x(t)=3\cos 3t+2\sin 3t+\frac{1}{3}t\sin 3t-\frac{1}{3}\cos 3t+\frac{1}{9}\ln\cos 3t\cdot\cos 3t$$

Seminar 13
27 mai 2014.

Transformarea Laplace

Transformarea „Z"

<u>Ex1</u> Să se rezolve ecuația integro-diferențială de tipul convoluției

$$x''(t)+2x'(t) = 2\int_0^t x'(\tau)\sin(t-\tau)\,d\tau + \cos t$$

$x'(0)=2$
$x(0)=0$

$$\boxed{(f*g)(t) = \int_0^t f(x)\cdot g(t-x)\,dx}$$

$$x''(t)+2x'(t) = 2(x'(t) * \sin t) + \cos t \quad \Big|\ \mathcal{L}$$

$$\mathcal{L}\{x''(t)\}(s) + 2\,\mathcal{L}\{x'(t)\}(s) = 2\mathcal{L}\{x'(t)*\sin t\}(s) + \mathcal{L}\{\cos t\}(s)$$

$$s^2X(s) - sx(0) - x'(0) + 2(sX(s)-x(0)) = 2\,\mathcal{L}\{x'(t)\}(s)\cdot\mathcal{L}\{\sin(t)\}(s) + \frac{s}{s^2+1}$$

$$s^2X(s) - 2 + 2sX(s) = 2sX(s)\cdot\frac{1}{s^2+1} + \frac{s}{s^2+1}$$

$$X(s)\left[s^2 + 2s + \frac{2s}{s^2+1}\right] = \frac{s}{s^2+1} + 2$$

$$X(s) = \frac{s+2s^2+2}{\cancel{s^2+1}}\cdot\frac{\cancel{s^2+1}}{s^4+s^2+2s^3+2s+2s}$$

$$X(s) = \frac{2s^2+s+2}{s^4+2s^3+s^2+4s}$$

$$= \frac{2s^2+s+1}{s^2(s+1)^2}$$

$\Rightarrow 2s^2+s+2 = As(s+1)^2 + B(s+1)^2 + Cs^2(s+1) + Ds^2$

$\Rightarrow B=2$

$A+C=0 \Rightarrow A=-3$

$D=0.$ $\quad C=3$

$\Rightarrow$

$X(s) = -\frac{3}{s} + \frac{2}{s^2} + \frac{3}{s+1} + \frac{3}{(s+1)^2}$

$x(t) = -3 + 2t + 3e^{t} + 3te^{-t}$

Ex 2 Se consideră funcția exponențial integrală definită astfel: $E_i(t) = \int_t^\infty \frac{e^{-x}}{x}\,dx \,;\; t > 0$

(i) să se calculeze transformata Laplace

(ii) să se calculeze:

$$\int_0^\infty t e^{-2t} \cdot E_i(t)\,dt = ?$$

Rezolvare:

$$\mathcal{L}\{E_i(t)\}(s) = \int_0^\infty E_i(t) \cdot e^{-st}\,dt$$

$$= \int_0^\infty \left[\int_t^\infty \frac{e^{-x}}{x}\,dx \right] e^{-st}\,dt$$

$$= \int_0^\infty e^{-st}\,dt \int_t^\infty \frac{e^{-xt}}{x}\,dx$$

$$\xrightarrow[dx = t\,dy]{x = ty} \int_0^\infty e^{-st}\,dt \cdot \int_1^\infty \frac{e^{-yt}}{y\cancel{t}} \cdot \cancel{t}\,dy =$$

$$= \int_0^\infty e^{-st}\,dt \int_1^\infty \frac{e^{-yt}}{y}\,dy$$

$$= \int_1^\infty \frac{1}{y}\,dy \int_0^\infty e^{-st} \cdot e^{-yt}\,dt$$

$$= \int_1^\infty \frac{1}{y}\,dy \int_0^\infty e^{-t(s+y)}\,dt$$

$$= \int_0^\infty \frac{1}{y} \cdot e^{\dots} \frac{}{-(s+y)} \Big|_0 \, dy$$

$$= \int_1^\infty \frac{1}{y} \cdot \frac{-1}{(1+y)} \left(\frac{e^{-\infty} - 1}{-1} \right) = \int_0^\infty \frac{1}{y(1+y)} \, dy$$

$$= \frac{1}{s} \int_0^\infty \frac{(s+y) - y}{y(s+y)} \, dy = \int_1^\infty \left(\frac{1}{y} - \frac{1}{s^2+y} \right) dy$$

$$= \frac{1}{s} \left(\ln y - \ln(1+y) \right) \Big|_1^\infty$$

$$= -\frac{1}{s} \ln \frac{s}{s+1} = \frac{1}{s} \ln(s+1)$$

(ii) $\int_0^\infty t \cdot e^{-2t} \cdot E_i(t)\, dt = \mathcal{L}\{ t \cdot E_1(t) \}(-2)$

$$\mathcal{L}\{ f(t) \}(s) = \int_0^\infty \left(f(t) e^{-st} \, dt \right) = (-1)^1 \left(\mathcal{L}\{ E_1(t) \}(s) \right)' \Big|_{s=2} =$$

$$= -\left(\frac{\ln(s+1)}{s} \right)'_{s=2} = -\left[\frac{\frac{1}{s+1} - \ln(s+1)}{s^2} \right]_{s=2} = \frac{3\ln 3 - 2}{2}.$$

Ex3 Să se calculeze suma:

$$s = \sum_{n=1}^\infty \frac{2n^2(-1)^n + n \cdot 2^n + 4\cos^2\left(\frac{n\pi}{3}\right)}{3^n}$$

$$\mathcal{Z}\{ x(n) \}(z) = \sum_{n=0}^\infty \frac{x(n)}{z^n} \; ; \; x \in S_d^+$$

$$\begin{cases} x(n) = 2n^2(-1)^n - n\,2^n + 4\cos^2\left(\frac{n\pi}{3}\right) \\ z = 3 \end{cases}$$

$$s = Z\{x(n)\}(3) - \frac{x(0)}{3^0}$$

$$Z\{2n^2(-1)^n\}(3) + Z\{n\cdot 2^n\}(3) + Z\{4\cos^2\left(\frac{n\pi}{3}\right)\}(3) - 4$$

$$= 2Z\{n^2\}(-3) \cdot \frac{2z}{(z-2)^2}\bigg|_{z=3} + \frac{4}{2} Z\{1+\cos^2\frac{2n\pi}{3}\}(3) - 4$$

$$Z\{a^n u(n)\}(z) = Z\{x(n)\}\left(\frac{z}{a}\right)$$

$$= 2\frac{z(z+1)}{(z-1)^3}\bigg|_{z=-3} + 6 + 2\cdot\frac{z}{z-1}\bigg|_{z=3} + 2\cdot\frac{z\left(z-\cos\frac{2\pi}{3}\right)}{z^2-2z\cos\frac{2\pi}{3}-1}\bigg|_{z=3} - 4$$

$$= 2\cdot\frac{6}{-64} + 6 + \cancel{2}\cdot\frac{3}{\cancel{2}} + 2\cdot\frac{3\left(3+\frac{1}{2}\right)}{9-6\left(-\frac{1}{2}\right)+1} - 4$$

$$= -\frac{3}{16} + 5 + \frac{21}{13}$$

Ex4 Să se calculeze:

$$s = \sum_{n=-1}^{\infty} \frac{n\cdot\sin^2\left(\frac{n\pi}{4}\right)}{2^n}$$

$$\begin{cases} x(n) = n\sin^2\left(\frac{n\pi}{4}\right) \\ z = 2 \end{cases}$$

$$s = Z\{n\sin^2\frac{n\pi}{4}\}(2) + \frac{x(-1)}{2^{-1}}$$

$$= \frac{1}{2}\{n\left(1-\cos\frac{n\pi}{2}\right)\}(2) - \frac{1}{2}\cdot 2 = \frac{1}{2}Z\{n\}(2) - \frac{1}{2}Z\{n\cos\frac{n\pi}{2}\} - 1 =$$

$$= \frac{1}{2}\frac{z}{(z-1)^2}\bigg|_{z=2} + \frac{1}{2}z\left(Z\{\cos\frac{n\pi}{2}\}(z)\right)'\bigg|_{z=2} - 1$$

$$= \frac{1}{2}\cdot 2 + \frac{1}{2}\cdot\left(\frac{z\left(z-\cos\frac{\pi}{2}\right)}{z^2-2z\cos\frac{\pi}{2}+1}\right)'\Bigg|_{z=2}$$

$$= \frac{1}{2}\left(\frac{z^2}{z^2+1}\right)'\Bigg|_{z=2} = \frac{1}{2}\left[\frac{2z(z^2+1)-z^2\cdot 2z}{(z^2+1)^2}\right]\Bigg|_{z=2}$$

$$= \frac{1}{2}\,\frac{2z}{(z^2+1)^2}\Bigg|_{z=2} = \frac{2}{(4+1)^2} = \frac{2}{25}.$$

EX5 Să se rezolve ecuația cu diferențe finite:

$$Z\Big|\; x(n+2)+2x(n+1)-8x(n)=2^n \;;\; n\geq 0$$

$x(0)=0$

$x(1)=3$.

Obs $x_n = x(n)$

$$\Rightarrow x_{n+2}+2x_{n+1}-8x_n = 2^n \;;\; n\geq 0$$

$x_0 = 0$

$x_1 = 3$

$$Z\{x(n+2)\}(z) + 2Z\{x(n+1)\}(z) - 8Z\{x(n)\}(z) = Z\{2^n\}(z)$$

notam $X(z) = Z\{x(n)\}(z)$

$$\Rightarrow \boxed{x(n) = Z^{-1}\{X(z)\}(z) \quad (1)}$$

$$Z\{x(n+1)\}(z) = zX(z) - z\cdot x(0) \;\Big|\; z$$

$$Z\{x(n+2)\}(z) = z^2X(z) - z^2x(0) - zx(1)$$

$$(z^2X(z) - z^2\cdot 0 - z\cdot 3) + 2(zX(z) - z\cdot 0) - 8X(z) = \frac{z}{z-2}$$

$$X(z)\left[z^2+2z-8\right] = \frac{z}{z-2} + 3z = \frac{z+3z^2-6z}{z-2}$$

$$X(z) = \frac{3z^2-5z}{(z-2)(z^2+2z-8)}$$

$$X(z) = \frac{z(3z-5)}{(z-2)^2(z+4)} \quad \Big| \; Z^{-1}$$

$$\frac{X(z)}{z} = \frac{3z-5}{(z-2)^2(z+4)}$$

$$3z-5 = A(z-2)(z+4) + B(z+4) + C(z-2)^2$$

$A = \frac{17}{36}$

$B = \frac{1}{6}$

$C = -\frac{17}{36}$

$$\Rightarrow \frac{X(z)}{z} = \frac{17}{36(z-2)} + \frac{1}{6(z-2)^2} - \frac{17}{36(z+4)} \qquad (2)$$

$$X(z) = \frac{17}{36} \cdot \frac{z}{z-2} + \frac{1}{6} \cdot \frac{z}{(z-2)^2} - \frac{17}{36} \cdot \frac{z}{z+4} \quad \Big| \; Z^{-1}$$

$$\overset{(1)}{\Rightarrow} x(n) = \left[\frac{17}{36} \cdot 2^n + \frac{1}{6} n \cdot 2^{m-1} - \frac{17}{36} \cdot (-4)^n\right] \cdot u(n)$$

cam $n \geq 0$.

$x(0) = 0 \rightarrow A''$

$x(1) = 3 \rightarrow A''$

1°. Să se calculeze transformata în Z:

$$x(n) = \frac{(-1)^n}{2n+1}\ ;\ x \in S_d^+$$

$$Z\{x(n)\}(z) = \sum_{n=0}^{\infty} \frac{x(n)}{z^n} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)z^n}$$

$$(\text{arctg}\, z)' = \frac{1}{1+z^2} = \sum_{n=0}^{\infty} (-1)^n z^{2n}\ ;\ |z^2|<1 \Rightarrow |z|<1$$

$$\text{arctg}\, z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{2n+1} + C\ ;\ |z|<1$$

$$z=0 \Rightarrow 0 = \sum_{n=0}^{n} (-1)^n \frac{z^{2n+1}}{2n+1} + C \Rightarrow C=0$$

$$\text{arctg}\, z = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot z^{2n+1}}{2n+1}\ ;\ |z|<1$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)} \cdot \left(\frac{1}{\sqrt{z}}\right)^{2n+1} \cdot \sqrt{z}$$

$$= \sqrt{z}\ \text{arctg} \frac{1}{\sqrt{z}}$$

De întrebat!

$$\underline{Y = L(x) = h * x}$$

$$h = \delta_{-2}^{1} + 2\delta_{n-1} - 3\delta$$

2°. $x(n) = \frac{n^2+1}{2^n \cdot n!}\ ;\ x \in S_d^+$

$$Z\{x(n)\}(z) = \sum_{n=0}^{\infty} \frac{x(n)}{z^n} = \sum_{n=0}^{\infty} \frac{n^2+1}{2^n n! \cdot z^n}$$

$$= \sum_{n=0}^{\infty} \frac{n(n-1)+n+1}{2^n \cdot n! \cdot z^n}$$

$$= \sum_{n=0}^{\infty} \frac{1}{2^n n! \cdot z^n} + \sum_{n=1}^{\infty} \frac{1}{2^n (n-1)! z^n} + \sum_{n=2}^{\infty} \frac{1}{2^n (n-2)! z^n}$$

$$= \sum_{n=0}^{\infty} \frac{1}{n!} \cdot \left(\frac{1}{2z}\right)^n + \frac{1}{2z} \sum_{n=0}^{\infty} \frac{1}{n!} \left(\frac{1}{2z}\right)^n + \frac{1}{4z^2} \sum_{k=0}^{\infty} \frac{1}{k!} \left(\frac{1}{2z}\right)^k$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$=\left(1+\frac{1}{2z}+\frac{1}{4z^2}\right)e^{\frac{1}{2z}}$$

$$\sum_{n=1}^{\infty}\frac{n^2+2n+5}{3^n}=\underbrace{\sum_{n=0}\frac{n^2+2n+5}{3^n}-5}_{}$$

$$\mathcal{Z}\{n^2+2n+5\}(3)$$

$$\mathcal{Z}\{n^2+2n+5\}(z)=\mathcal{Z}\{n^2\cdot u(n)\}(z)+\mathcal{Z}\{2n\cdot u(n)\}(z)+\mathcal{Z}\{5u(n)\}(z)$$

$$=\frac{z(z+1)}{(z-1)^3}+2\frac{z}{(z-1)^2}+\frac{5z}{z-1}$$

$$\Rightarrow\sum_{n=1}^{\infty}\frac{n^2+2n+5}{3^n}=\frac{3\cdot 4}{2^3}+2\cdot\frac{3}{2^2}+\frac{5\cdot 3}{2}-5$$

$$=\frac{12}{8}+\frac{3}{2}+\frac{15}{2}-5$$

$$=-2+\frac{15}{2}=\frac{11}{2}$$

(i) $$\sum_{n=0}^{\infty}\frac{(-1)^n(2n^2+7)}{3^n}=\mathcal{Z}\{(-1)^n(2n^2+7)\}(3)$$

(ii) $$\sum_{n=2}^{\infty}\frac{(-1)^n\cdot n^2+\cos\frac{n\pi}{3}}{2^n}=\mathcal{Z}\left\{(-1)^n\cdot n^2+\cos\frac{n\pi}{3}\right\}(2)-1+\cancel{1}-\cos\frac{\pi}{3}$$

$$=\mathcal{Z}\left\{(-1)^n\cdot n^2+\cos\frac{n\pi}{3}\right\}(2)$$

(iii) $$\sum_{n=1}^{\infty}\frac{n\sin\frac{n\pi}{6}}{3^n}$$

$$\mathcal{Z}\left\{(-1)^n\cdot n^2+\cos\frac{n\pi}{3}\right\}(z)=-\frac{z(z-1)}{(z+1)^3}+\frac{z\left(z-\cos\frac{\pi}{3}\right)}{z^2-2z\cos\frac{\pi}{3}+1}$$

$$\mathcal{Z}\{a^n\cdot u(n)\}(z)=F\left(\frac{z}{a}\right)$$

$a=-1$

$$\mathcal{Z}\{n^2\cdot u(n)\}(z)=\frac{z(z+1)}{(z-1)^3}$$

$$\mathcal{Z}\{(-1)^n\cdot n^2\cdot u(n)\}(z)=\frac{-z(-z+1)}{(-z-1)^3}=\frac{z(z-1)}{(z+1)^3}$$

$$= -\frac{z(z-1)}{(z+1)^3} + \frac{z(z-\frac{1}{2})}{z^2-z+1}$$

$$\sum_{n=2}^{\infty} \frac{(-1)^n n^2 + \cos\frac{n\pi}{3}}{2^n} = \frac{-2}{3^3} + \frac{2(2-\frac{1}{2})}{4-2+1} - \frac{1}{2}$$

$$= \frac{-2}{27} + 1 - \frac{1}{2}$$

$$= \frac{-4+54-27}{54} = \frac{23}{54}.$$

$$\sum_{n=1}^{\infty} \frac{n \sin\frac{n\pi}{6}}{3^n} = \sum_{n=0}^{\infty} \frac{n \sin\frac{n\pi}{6}}{3^n} = Z\left\{n \sin\frac{n\pi}{6}\right\}(3)$$

$$= -\frac{3}{2} \cdot \frac{1-9}{(9-3\sqrt{3}+1)^2} = \frac{12}{(10-3\sqrt{3})^2} = \frac{12}{100-60\sqrt{3}+27} = \frac{12}{127-60\sqrt{3}}$$

$$Z\left\{\sin\frac{n\pi}{6}\right\}(z) = \frac{z \sin\frac{\pi}{6}}{z^2-2z\cos\frac{\pi}{6}+1} = \frac{\frac{1}{2} z}{z^2-\cancel{2}z\cdot\frac{\sqrt{3}}{\cancel{2}}+1}$$

$$= \frac{z}{2(z^2-z\sqrt{3}+1)}$$

$$Z\{n \cdot f(n)\}(z) = -z F'(z)$$

$$= -\frac{z}{2}\left(\frac{z}{z^2-z\sqrt{3}+1}\right)'$$

$$= -\frac{z}{2} \cdot \frac{z^2-z\sqrt{3}+1-2z^2+z\sqrt{3}}{(z^2-z\sqrt{3}+1)^2}$$

$$Z^{-1}\left\{\frac{z^2+2z-6}{z-2}\right\}(n)$$

$$\frac{F(z)}{z} = \frac{z^2+2z-6}{z(z-2)} = 1 + \frac{4z-6}{z(z-2)} = 1 + \frac{3}{z} + \frac{1}{z-2}$$

$$\frac{4z-6}{z(z-2)} = \frac{3}{z} + \frac{1}{z-2}$$

$$\Rightarrow F(z) = \underset{\delta_{-1}}{\underset{\uparrow}{z}} + \underset{3\delta_0}{\underset{\uparrow}{3}} + \underset{2^n \cdot u(n)}{\underset{\uparrow}{\frac{z}{z-2}}} = \delta_{-1} + 3\delta_0 + 2^n \cdot u(n).$$

3°. Să se rezolve următoarele ecuații cu diferențe finite

$x \in \mathcal{S}^+$

$$x(n+2) + 3x(n+1) + 2x(n) = (-5)^n \Big| \cdot \mathcal{Z} \quad n \geq 0 \qquad \begin{array}{l} x(0)=2 \\ x(1)=-1 \end{array}$$

notăm:

$$X(z) = \mathcal{Z}\{x(n)\}(z)$$

$$\mathcal{Z}\{x(n+1)\}(z) = zX(z) - zx(0)$$
$$= zX(z) - 2z$$

$$\mathcal{Z}\{x(n+2)\}(z) = z^2X(z) - z^2x(0) - zx(1)$$
$$= z^2X(z) - 2z^2 + z$$

$$z^2X(z) - 2z^2 + z + 3\left(zX(z) - 2z\right) + 2X(z) = \frac{z}{z+5}$$

$$(z^2+3z+2)X(z) = \frac{z}{z+5} + 2z^2 - z + 6z$$

$$(z+1)(z+2)X(z) = \frac{z}{z+5} + 2z^2 + 5z = \frac{2z^3 + 5z^2 + 10z^2 + 25z + z}{z+5}$$

$$X(z) = \frac{2z^3 + 15z^2 + 26z}{(z+1)(z+2)(z+5)}$$

$$\frac{X(z)}{z} = \frac{2z^2 + 15z + 26}{(z+1)(z+2)(z+5)} = \frac{A}{z+1} + \frac{B}{z+2} + \frac{C}{z+5}$$

$$A = \frac{2z^2 + 15z + 26}{(z+2)(z+5)}\Bigg|_{z=-1} = \frac{2 - 15 + 26}{1 \cdot 4} = \frac{13}{4}$$

$$B = \frac{2z^2 + 15z + 26}{(z+1)(z+5)}\Bigg|_{z=-2} = \frac{8 - 30 + 26}{(-1) \cdot 3} = \frac{4}{-3} = -\frac{4}{3}$$

$$C = \frac{2z^2+15z+26}{(z+1)(z+2)}\Big|_{z=-5} = \frac{50-75+26}{(-4)\cdot(-3)} = \frac{1}{12}.$$

$$\Rightarrow \frac{X(z)}{z} = \frac{13}{4}\cdot\frac{1}{z+1} - \frac{4}{3}\cdot\frac{1}{z+2} + \frac{1}{12}\cdot\frac{1}{z+5}$$

$$X(z) = \frac{13}{4}\cdot\frac{z}{z+1} - \frac{4}{3}\cdot\frac{z}{z+2} + \frac{1}{12}\cdot\frac{z}{z+5}$$

$$= \frac{13}{4}(-1)^n - \frac{4}{3}(-2)^n + \frac{1}{12}(-5)^n$$

4° $$x_{n+2} - 4x_{n+1} + 4x_n = 3^n \Big|_{Z} \; ; n \geq 0 \; ; \; x_0 = 0, \; x_1 = 2.$$

notăm:

$$X(z) = Z\{x_n\}(z) = Z\{x(n)\}(z)$$

$$Z\{x(n+1)\} = zX(z) - z x_0 = zX(z)$$

$$Z\{x_{n+2}\}(z) = z^2 X(z) - z^2 x_0 - z x_1$$
$$= z^2 X(z) - 2z$$

$$\{3^n \cdot u(n)\}(z) = \frac{z}{z-3}$$

$$z^2 X(z) - 2z - 4z X(z) + 4X(z) = \frac{z}{z-3}$$

$$(z^2 - 4z + 4)X(z) = 2z + \frac{z}{z-3} = \frac{2z^2 - 6z + z}{z-3} = \frac{2z^2 - 5z}{z-3}$$

$$X(z) = \frac{z(2z-5)}{(z-3)(z-2)^2}$$

$$\frac{X(z)}{z} = \frac{2z-5}{(z-3)(z-2)^2} = \frac{A}{z-3} + \frac{B}{z-2} + \frac{C}{(z-2)^2}$$

$A = 1$

$$B = \operatorname{Rez}\left(\frac{2z-5}{(z-3)(z-2)^2}; 2\right) = \lim_{z \to 2} (z-2)^2 \cdot \frac{2z-5}{(z-3)(z-2)^2} = -1$$

$C = 1$

$$\Rightarrow \frac{X(z)}{z} = \frac{1}{z-3} + \frac{1}{(z-2)^2} - \frac{1}{z-2}$$

$$x(n) = \frac{z}{z-3} - \frac{z}{z-2} + \frac{z}{(z-2)^2}$$

$$= 3^n - 2^n + n\,2^{n-1}$$

$$\mathcal{Z}\{n\,a^n\}(z) = \frac{az}{(z-a)^2}$$

5°. $x(n+2) \cdot x^2(n+1) = e\, x^3(n)$ $\Big|$; $n \geq 0$; $x_0 = x_1 = 1$.

$\ln$

$$\ln x(n+2) + 2\ln x(n+1) = 1 + 3\underbrace{\ln x(n)}_{y(n)}$$

$$y(0) = \ln x(0) = 0$$

$$y(1) = \ln x(1) = 0$$

$$\Rightarrow y(n+2) + 2y(n+1) = 1 + 3y(n)$$

$$y(n+2) + 2y(n+1) - 3y(n) = 1$$

$$Y(z) = \mathcal{Z}\{y(n)\}(z)$$

$$\mathcal{Z}\{y(n+1)\}(z) = zY(z) - zy(0) = zY(z)$$

$$\mathcal{Z}\{y(n+2)\}(z) = z^2Y(z) - z^2y(0) - zy(1) = z^2Y(z)$$

$$\mathcal{Z}\{1\}(z) = \mathcal{Z}\{u(n)\}(z) = \frac{z}{z-1}$$

$$z^2Y(z) + 2zY(z) - 3Y(z) = \frac{z}{z-1}$$

$$(z^2 + 2z - 3)Y(z) = \frac{z}{z-1}$$

$$\frac{Y(z)}{z} = \frac{1}{(z-1)^2(z+3)} = \frac{A_1}{z-1} + \frac{A_2}{(z-1)^2} + \frac{B}{z+3} = -\frac{1}{16}\cdot\frac{1}{z+1} + \frac{1}{4}\cdot\frac{1}{(z-1)^2} + \frac{1}{16}\cdot\frac{1}{z+3}$$

$$A_2 = \frac{1}{z+3}\Big|_{z=1} = \frac{1}{4}$$

$$B = \frac{1}{(z-1)^2} = \frac{1}{16} \Rightarrow A_1 = -\frac{1}{16}.$$

$$y(z) = -\frac{1}{16}(-1)^n + \frac{1}{4}\cdot n + \frac{1}{16}$$

$$x(n) = e^{\left(-\frac{1}{16} + \frac{n}{4} + \frac{(-3)^n}{16}\right)} \cdot u(n)$$